

BUSINESS SALES PERFORMANCE ANALYTICS

Superstore Dataset | 2014 – 2017

Tools Used: SQL Server Management Studio (SSMS) & Microsoft Excel
Track: Data Science & Analytics | Repository: FUTURE_DS_01

KEY PERFORMANCE INDICATORS

\$2.30M Total Revenue All 4 regions	\$286.8K Total Profit 12.5% margin	West Top Region \$725K / 14.9%	Technology Top Category \$836K / 17.4%
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1. Executive Summary

This report presents a comprehensive analysis of the Superstore retail dataset spanning four fiscal years (2014–2017). Using SQL queries in SSMS for data extraction and aggregation, and Microsoft Excel for visualization, the analysis covers four dimensions: regional performance, monthly sales trends, product category profitability, and the impact of discount strategies on profit margins. Key findings reveal strong revenue growth year-over-year, concentrated profitability in two regions, and a critical pricing risk introduced by aggressive discounting above 20%.

2. Regional Performance

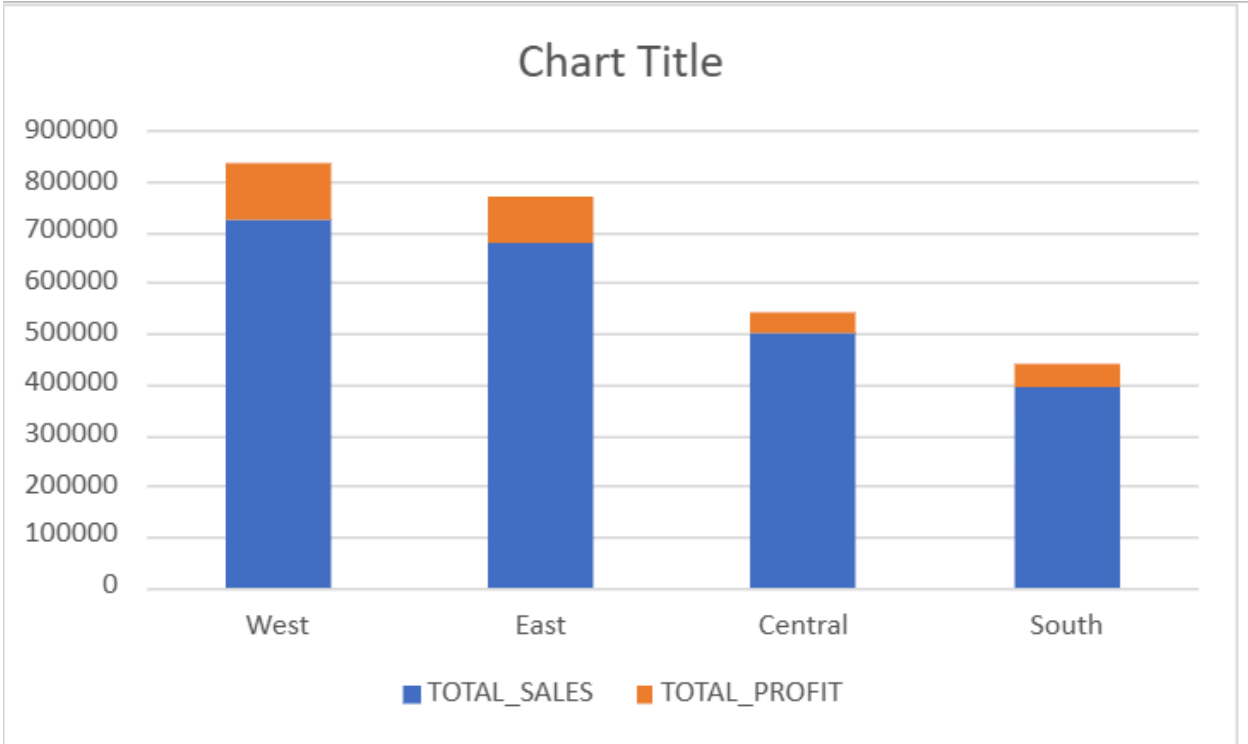
2.1 Revenue & Profit by Region

The West region led all four regions in both revenue and profitability, while the Central region showed the weakest profit margin despite ranking third in revenue — indicating potential cost or discount issues in that market.

Region	Total Revenue	Total Profit	Margin %	Rank
West	\$725,457	\$108,418	14.9%	🥇 1st
East	\$678,781	\$91,523	13.5%	2nd
Central	\$501,240	\$39,706	7.9%	3rd
South	\$391,722	\$47,169	12.0%	4th

TOTAL	\$2,297,200	\$286,816	12.5%	—
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Table 1: Regional revenue, profit, and margin summary (2014–2017)



2.2 Regional Insights

- West and East together account for 61% of total revenue and 69% of total profit.
- Central region's margin of 7.9% is significantly below the company average — warrants a pricing or cost audit.
- South region punches above its revenue weight with a healthy 12.0% margin.

3. Monthly Sales Trend (2014–2017)

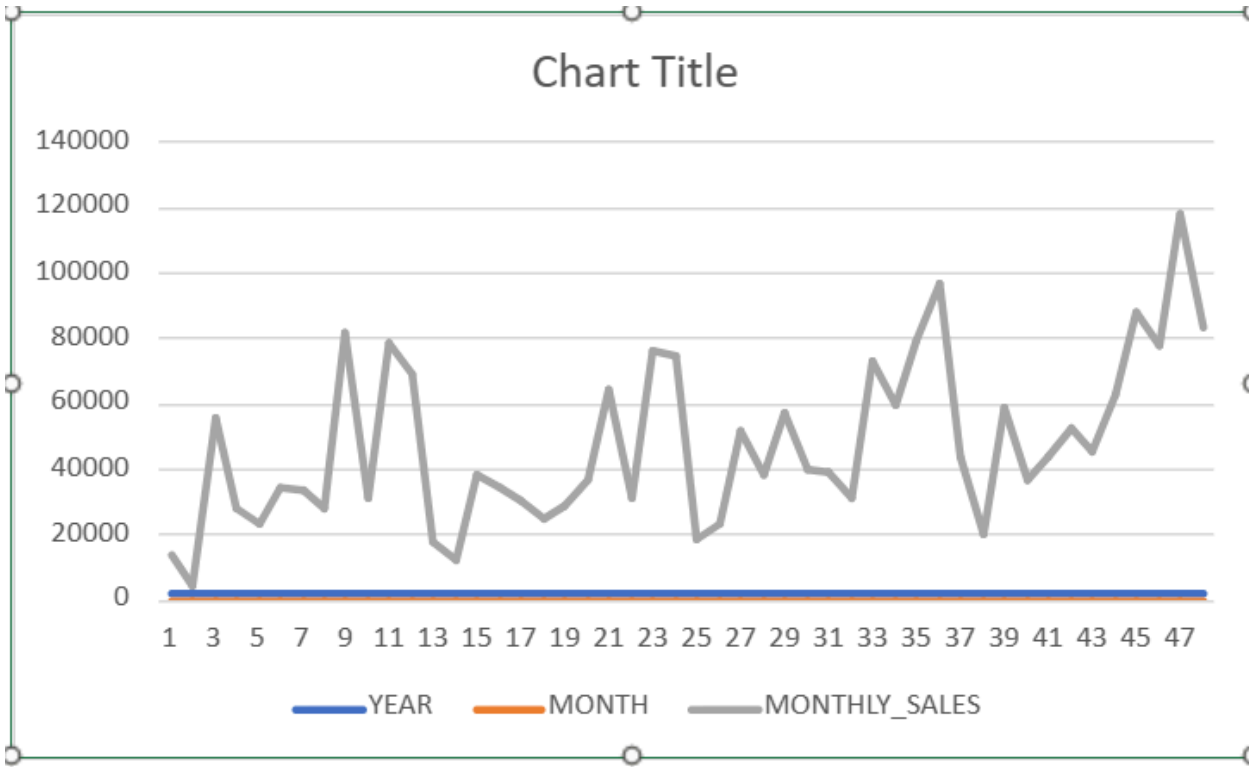
3.1 Year-over-Year Growth

Total annual sales grew consistently across all four years, with 2017 recording the highest revenue at approximately \$733K — an 81% increase over 2014's \$484K baseline.

Year	Annual Sales	YoY Growth	Peak Month
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2014	\$484,247	Baseline	Sep (\$81.8K)
2015	\$470,533	-2.8%	Nov (\$76.0K)
2016	\$609,206	+29.5%	Dec (\$97.0K)
2017	\$733,215	+20.4%	Nov (\$118.4K)

Table 2: Annual sales summary with YoY growth and peak month



3.2 Seasonal Patterns

- September and November are consistently the strongest sales months across all years — likely driven by back-to-school and end-of-year procurement cycles.
- Q1 (Jan–Feb) is the weakest quarter every year, with February averaging under \$15K — a potential window for targeted promotions.
- 2015 saw a slight revenue dip vs 2014, likely due to pricing adjustments or a temporary demand drop.

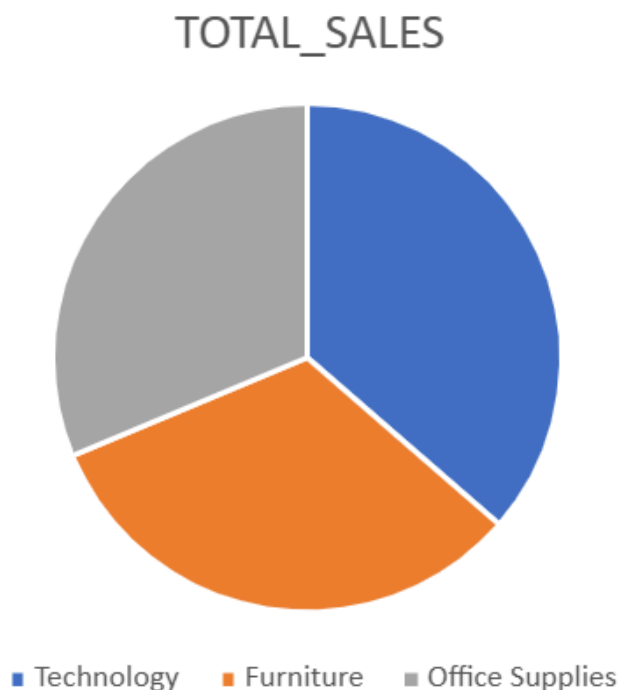
4. Product Category Analysis

4.1 Revenue vs. Profit by Category

Technology leads in both revenue and profit margin, making it the most valuable category. Furniture generates the second-highest revenue but has an extremely thin margin of just 2.5% — barely profitable, suggesting heavy discounting or high cost-of-goods. Office Supplies is the most efficiently profitable category relative to its revenue.

Category	Revenue	Profit	Margin %	Efficiency
Technology	\$836,154	\$145,455	17.4%	High
Furniture	\$742,000	\$18,871	2.5%	Low
Office Supplies	\$719,047	\$122,491	17.0%	High

Table 3: Category-level revenue, profit, and margin



- Furniture's 2.5% margin is a red flag — investigate sub-categories (likely Tables/Bookcases driving losses).
- Technology and Office Supplies together deliver 94% of total company profit on only 68% of revenue.
- Recommend reducing discounts on Furniture, especially in the Central region where both factors overlap.

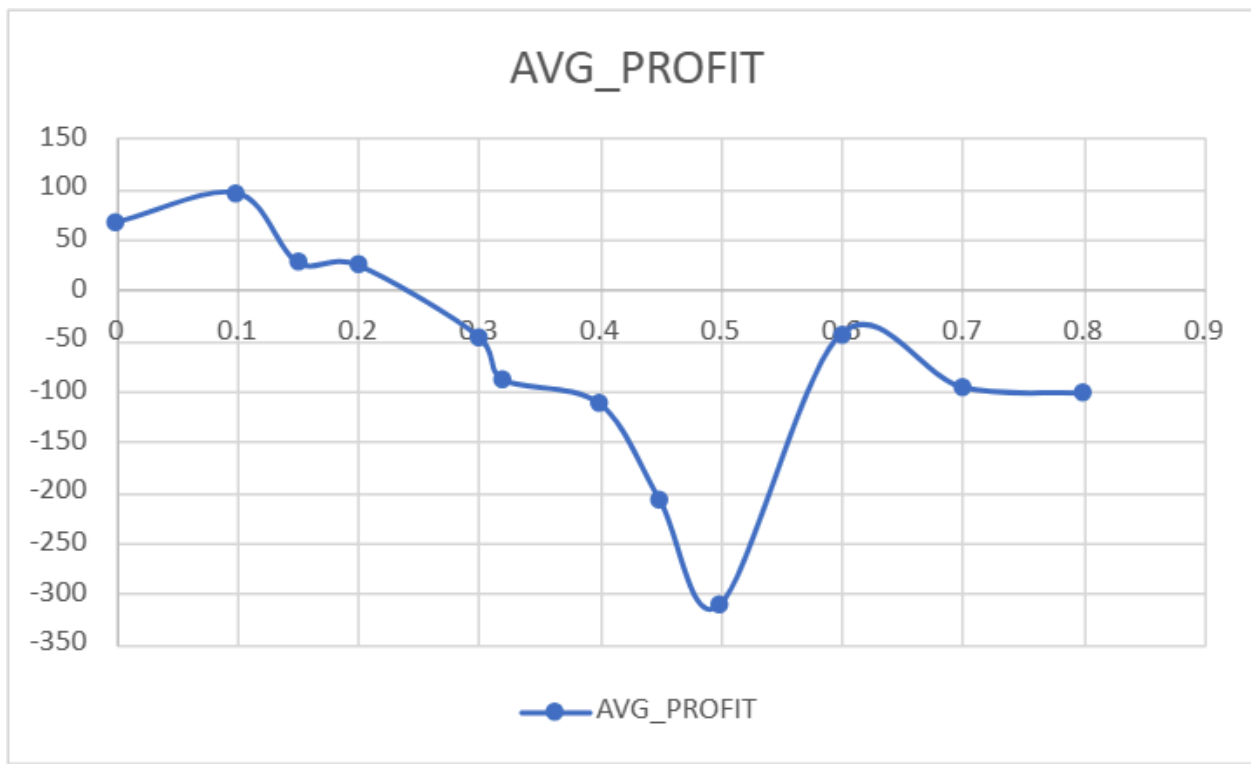
5. Discount Strategy & Profit Impact

5.1 Discount Rate vs. Average Profit

This is the most actionable finding in the dataset. Average profit is positive at discount levels of 0%–20% but turns sharply negative beyond 30%. Extreme discounts (45%–50%) result in average losses of \$200–\$310 per order.

Discount	Avg Profit (\$)	Status	Recommendation
0%	\$66.90	Profitable	Maintain — healthy baseline profit
10%	\$96.06	Optimal	Best profitability — sweet spot discount
15%–20%	\$25–\$27	Marginal	Use cautiously — thinning margins
30%+	-\$46 to -\$311	Loss-making	Cap discounts at 20% — immediate action needed

Table 4: Discount rate impact on average profit per order



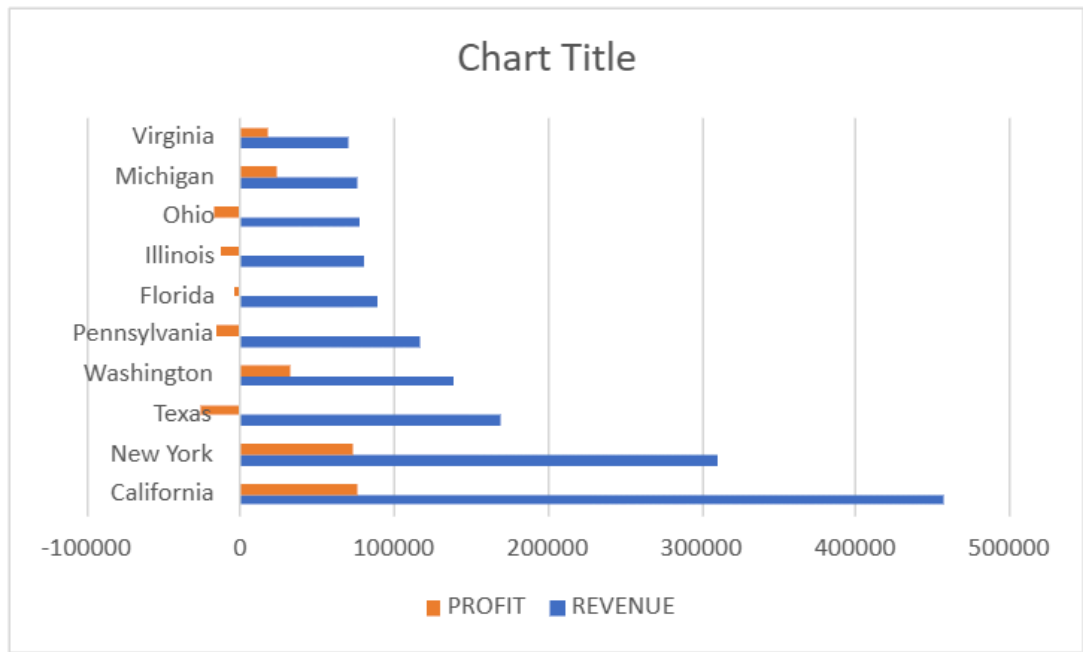
- 10% discount is paradoxically the highest-profit point — customers spending more without expecting deep cuts.
- Every order with a 30%+ discount is, on average, destroying value. This policy needs urgent review.
- The 60%–80% discount band shows less extreme losses — likely small orders where the absolute dollar impact is limited.

6. State-Level Performance (Top 10)

California and New York are the top two states by revenue and are also among the most profitable. Texas, despite being the third-highest revenue state at \$170K, is generating a loss of -\$25.7K — the worst in the dataset. Several mid-tier states also run negative profit.

State	Revenue	Profit	Margin %	Flag
California	\$457,688	\$76,381	16.7%	Star market
New York	\$310,876	\$74,039	23.8%	Highest margin
Texas	\$170,188	-\$25,729	-15.1%	Urgent review needed
Washington	\$138,641	\$33,403	24.1%	Strong performer
Pennsylvania	\$116,512	-\$15,560	-13.4%	Unprofitable
Florida	\$89,474	-\$2,979	-3.3%	Marginally negative
Illinois	\$80,166	-\$12,608	-15.7%	Unprofitable
Ohio	\$78,258	-\$16,971	-21.7%	Worst margin in top 10
Michigan	\$76,270	\$24,463	32.1%	Top margin state
Virginia	\$70,637	\$18,598	26.3%	Solid performer

Table 5: Top 10 states by revenue with profitability assessment



7. Actionable Recommendation

#	Area	Recommendation	Impact
1	Discount Policy	Cap discounts at 20% across all categories. Any deal above 20% requires manager approval.	High — stops value destruction immediately
2	Furniture Margin	Audit Furniture sub-categories (Tables likely loss-making). Review supplier costs and pricing.	High — 2.5% margin is unsustainable
3	Texas & Ohio	Investigate discount rates and product mix in Texas and Ohio, which are posting significant losses.	High — combined -\$43K loss
4	Q1 Promotions	January–February is consistently weak. Launch targeted promotions for Technology and Office Supplies to smooth seasonality.	Medium — revenue opportunity
5	Scale West Model	Replicate the West region's customer engagement and pricing strategy in the Central region.	Medium — close 7% margin gap

Table 6: Priority recommendations with expected business impact

8. Tools & Methodology

Tool	Usage
SSMS (SQL Server)	Data import, cleaning, and all SQL aggregation queries — regional sums, monthly groupings, category breakdowns, discount bucketing, state-level profit analysis.
Microsoft Excel	Pivot tables for cross-tabulation, bar and line charts for trend visualization, conditional formatting for highlighting loss-making segments.
Dataset	Sample Superstore (9,994 rows, 21 columns) — orders from Jan 2014 to Dec 2017 across 4 regions, 49 states, 3 product categories.

Table 7: Tools and data sources

9. Conclusion

The Superstore dataset reveals a business with strong overall growth but significant profit leakage concentrated in three areas: aggressive discounting above 20%, the Furniture category's near-zero margins, and specific state markets (Texas, Ohio, Illinois, Pennsylvania) running at a loss. Addressing just the discount policy alone could recover tens of thousands of dollars in annual profit. The West and East regions, combined with Technology and Office Supplies, demonstrate what a healthy margin profile looks like — and provide a blueprint for improving the underperforming segments.